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Chapter 6

Tips And Techniques



How To Use Tips And Techniques



Tip: This is a dynamically changing document incorporating comments and ideas from JBA consultants and customers in the field. If you have anything to add to this please contact JBA Software Products to help us keep the details as comprehensive and topical as possible.

When To Use Tips And Techniques

Use Tips And Techniques to resolve problems encountered with the System 21 interfaces.

Tips and Techniques refers to all these three interfaces, unless an individual interface is named specifically within the text:

- System 21 Client Workspace
- System 21 Explorer
- System 21 Constructor.

What Does Tips And Techniques Give You

It gives you:

- Troubleshooting help.
- Checklists.
- Detailed background information on various subjects such as the routers, the error codes, the configuration files shipped with the product.
- Useful suggestions for making further use of the product. For example how to carry out a trace.

Important Tips



Tip: After any failure, reboot the PC before attempting to run the System 21 interface again.



Tip: If a problem occurs on a setup which previously caused no problems, focus any efforts on the areas which may have changed.



Tip: If a problem occurs before the logo panel appears, check for missing objects such as path errors or router issues.

Troubleshooting



Note: These are the most common issues that have been encountered to date. They may all be resolvable without software changes.

All Tasks Present Sign On Panel

This occurs as a result of one of three situations:

1. The System 21 interface is detecting a message being returned by the Sign On panel. The message will appear on the redisplayed screen, and should be actioned accordingly.
2. The System 21 interface needs to be able to enter an Initial Program. It is set to the startup program CWSInitial
3. The above file holds what it will interpret as the Sign On panel, by storing the first word that appears on that panel. It is shipped as defaulting to IBM English Standard, but if customised, or in a foreign language, may need changing first. See SELECTHOST section of the configuration file for more details.

Application Error With “Cannot open”, “Cannot find”, “Cannot read”

Check the Autoexec.bat PATH and the location of the missing object.

Check the Config.sys file has SHARE loaded.

Application Message Queue Full

Please refer to Win.ini checks.

Arial Monospaced Font Not Found

This is caused by a font having been previously uninstalled incorrectly. The installation process then cannot install the new font as it expects to and reports that the font is not found.

The way around this problem is to manually install the font to the Fonts directory.

See also: [Printing Incorrectly From Acrobat Viewer\(.pdf\)](#). There is a potential problem where the printer is not set up correctly to receive this font.

Cannot Start Task At This Time

Note the status bar indicating when a session or task is opening. This message most often occurs after a previous task has been requested, but has yet to reach the 'Starting task in session...' status. When the message is 'Opening session...', a task cannot be started. Once a session is available, the previous task is then started. At this point, the next task may be initiated.

Wait until the status bar changes from 'Opening session...' to 'Starting task in session....'. If in doubt, wait ten seconds and try again.

The wait time is dependent on the performance of the router or link to the Host, as well as the PC specification.

Comms Error - Mapping Service To Port Number (starting session) - UNIX Only

PC limitation on File Handles. Adjust MAXSESSIONS in Jbas21p.ini file.

If the Comms Error occurs after Logon then check the following:

- Logon Error Messages
- The Routers to see if they are active
- Host messages. Refer to the Checklists

Connection Errors

If the error is a failure to connect to the host, make sure that the host is working correctly.

If you are using TCP/IP

- Make sure that the CSSXLAT environment variable is set to the location of a valid Jbagtab.dat file.

If you are using TCP/IP to AS/400(JVN):

- Make sure that TCP/IP is running on the AS/400
- Make sure that TCP/IP has been correctly configured.

If you are using TCP/IP to AS/400 or Retrofit (JVN or JVT):

- Make sure that the host can be contacted over the network - type "ping hostname" from a DOS prompt.

If there is a DNS server:

- Make sure that the DNS server is running and has the details for the host to which you are trying to connect
- Make sure that the jqserv service on the host has been mounted on the correct port. You can assume that the port is correct if another PC has made a successful connection.

If you are using a Router (JVW):

- Make sure that the router is connected to the AS/400
- Make sure that the devices on the AS/400 for the PC, for example: PcNameJ1, are varied on. You can try varying them off and on again.

A TCP/IP connection to an AS/400 might fail for the following reasons:

- Attempt to connect from a PC where TCP/IP is not correctly configured.
- Attempt to connect to a machine whose IP address cannot be found (if there is a problem with the DNS or the IP address is missing from the PC's hosts file).
- Operating system turned down request for TCP/IP connection (socket) - possibly running too many other applications.
- Attempt to connect to a machine which does not respond over the network. Make sure that TCP/IP is running on the host machine.
- Attempt to connect to a machine which is not an AS/400 - check the PC host configuration.
- Attempt to connect to an AS/400 which does not have TCP/IP started. Make sure that TCP/IP is running.
- Negotiate with AS/400 failed - PC and AS/400 could not agree on terminal type, end of record formats, binary transmission or Double Byte terminal type (for Double Byte hosts only). Make sure the AS/400 is correctly configured for TCP/IP and that there are TCP/IP devices. Use WRKDEVD DEVD(QPADEV*) to check the device descriptions for the AS/400 devices. The PC uses devices of type 3179.

For more information please refer to Checklists.

Designer Authorised But Does Not Appear

Check:

- If designer is switched on in Jbas21p.ini. Set Reformatting ON
- If you have two or more sessions open - you can only have a single session with Designer.
- If the current session has a magic number. It must have a magic number for designer to work.

Also, if you installed via a Network, check for an AUTHORITY problem on the server drive where the SR files are resident.

Designer Fails To Open SR Files

Check PC Virtual Memory setting.



Warning: 32 bit file and disk access must be off.



Warning: If the SR files were loaded onto a server drive, make sure that there is sufficient access authority to that drive.

Document Viewer Fails To View Web Pages

Check that you have correctly installed Microsoft Internet Explorer, version 3 or later. This is a Web browser which the Document Viewer needs to view Web pages.

Explorer Hangs



Note: This is applicable to AS/400 using TCP/IP.

If Explorer hangs after 5 to 10 minutes it is probably because of a configuration problem. There are two things you can try:

- Make sure that there is not a power saver set up on the PC. If the monitor switches itself into standby or if the hard disk stops spinning then this is the cause of the problem.
- Else it is probably an AS/400 configuration setting. Sign on as QSECOFR and use CHGTELNA to check the values of Inactivity timeout and Timemark timeout. Note that TCP/IP must be restarted on the AS/400 if values are changed.

This is how the screen should look:

Change TELNET Attributes (CHGTELNA)

Type choices, press Enter.

Autostart server	*YES	*YES, *NO, *SAME
Inactivity timeout	0	0-2147483647, *SAME, *DFT
Timemark timeout	600	0-2147483647, *SAME, *DFT
Default NVT type	*VT100	*SAME, *VT100, *NVT
Coded character set identifier	*MULTINAT	1-65533, *SAME, *MULTINAT...
ASCII fullscreen mapping:		
Outgoing EBCDIC/ASCII table .	*CCSID	Name, *SAME, *CCSID, *DFT
Library		Name, *LIBL, *CURLIB
Incoming ASCII/EBCDIC table .	*CCSID	Name, *SAME, *CCSID, *DFT
Library		Name, *LIBL, *CURLIB

Failed To Query Router

Router problem - cannot communicate with host.

If this message displays whilst attempting to start the System 21 interface itself, you must allow a short interval between closing down one System 21 interface session and starting another. Five seconds is normally enough.

If this was the first entry to the System 21 interface, then try again, after exiting Windows.

General Protection Fault In G21AWB3D.DLL Or In CSPE400W.DLL

Likely to be memory related problems.

Grey Screen In Designer Mode

Check for sufficient message queue requirements. See Win.ini checks.

Incorrect Characters Appear Consistently (Translation)

Check conversion requirements for Jbas21p.ini file in the OEMCONV section.

Check known router translation issues.

See also: Countries and Character Translation.

Insufficient Resources To Start Application Or Session

Close down all other tasks and sessions on the PC to free up resources

Lower the resource limit for the application or the session on the Preferences dialogue. This lowering will increase the chances of PC failure when its resource thresholds are reached.

This is a PC problem due to the limitations of the PC.

Logon Error Messages

If you encounter any of the following messages at logon:

- Cannot logon with the UserID
- Communications Error in Session
- Failed to create default session.

Please check the following:

1. Invalid supported setup. Check it is valid against the current certified support list.
2. Refer to the Checklists in Tips and Techniques.
3. Check for specific issues with Routers.
4. Run through the User Checks.
5. Run through other issues detailed in the Contents of this InfoBase (For example, Security, Other Emulators and Memory Issues).

If the error occurs whilst starting an additional session in the System 21 interface, then there may be caused by the limitations of the PC.

No Help For Error Messages

This may be because InfoBase has not been installed with Client Workspace and the configuration file is pointing to the wrong location. Please refer to the section called About The Help Files.

Out Of Environment Space

This may be resolved by changing the Config.sys file setup.

Printing Incorrectly From Acrobat Viewer(.pdf)

Potential Problem

JBA issue Help files in .pdf format, for example wm.pdf, for use with Acrobat Viewer. You can print pages directly from Acrobat Viewer, however, there may be a problem where Arial Monospaced font has been used to show excerpts of program code and the printer is not set up correctly to receive this font.

How To Resolve It

To print all fonts correctly on your printer check the following:

- That you set the PC to the correct print resolution of 300 dots per inch.
- That you only print from your hard drive or a network drive.

If you do not know how to do this then please take the following steps. You need only do this once:

1. Close Acrobat viewer.
2. Load the .pdf file onto your hard drive on the PC, or to a network drive. If the .pdf file has been sent to you via Lotus Notes then please detach to the hard drive.
3. Select Start\Settings\Printers to display the Printers Window.
4. Select your default printer and click with the right mouse button. This displays the pop up menu.
5. Select Properties from the pop up menu.
6. Select the panel with the Resolution field (usually the Graphic panel).
7. Change the Resolution to 300 dots per inch.
8. Click Apply and then click OK.
9. Open Acrobat Viewer and resume printing from your .pdf file as normal.

Program Not Launching But Installed OK



Tip: This may be an anti-virus software problem.

System 21 Interfaces use InstallShield as the installation software package. InstallShield recognise that some anti-virus programs can interfere with an installation. Different anti-virus programs offer different levels of protection and in most cases there will be no problem. However it is possible that your installation will look as if it is running smoothly, your icons will be created in Program Manager, your dialogue boxes will appear but the anti-virus program will not allow your program to launch.

This happens because most anti-virus programs do two things:

- Prevent you from copying EXE, DLL, COM and other program files and data to your hard disk.
- Watch for illegal operations; for example, any attempt to write to the boot sector of your hard disk.

The problem is that anti-virus programs cannot distinguish between harmful and helpful software so they can stop you from installing new software also.

The Central Point and MS-DOS Anti-virus programs offer more than one level of protection. At one level, the program will lock any new EXE program file being copied to the hard disk, and will not allow the program to run until it has been scanned for viruses. For most anti-virus programs, the default level of protection is much lower and does allow you to copy an EXE file to the hard disk.

Before InstallShield can install your software on the user's hard drive, it copies the setup program from the floppy disk to the hard disk, and then launches the setup from the hard disk. If an anti-virus program is running then there is no way InstallShield or any setup can override it.

This is what you can do:

Step 1

Open Autoexec.bat using a text editor.

Step 2

Go to the beginning of each line that starts an anti-virus program. There is usually only one line. Comment out the line by typing REM followed by a space.

Step 3

Save Autoexec.bat.

Step 4

Open Config.sys using a text editor.

Step 5

Go to the beginning of each line that starts an anti-virus program. There is usually only one line. Comment out the line by typing REM followed by a space.

Step 6

Save Config.sys.

Step 7

Reboot the computer and try the setup again.

Signon Screen Appears For UNIX, But Login Impossible

Check SVSIGNON.

*.SMU Not Found

Check the SMUPath in the Preferences dialogue, and the location of the missing .SMU

System 21 Interface Doesn't Start In UNIX

1. Check correct HostType in Jbas21p.ini, and [Hosts] Host0 details in section JBAVT of the same file. PC problem. Refer to First Host.
2. Check Configuration setup of Services and Inetd. on the PC and Host.
3. Check other General Checks (UNIX/AIX).

Translation Table Not Loaded

Check the SET CSSXLAT statement in Autoexec.bat and the location of file Jbagtab.dat.
Check location of SET statements in Autoexec.bat file.

Unexpected Panel Received During Logon

Pass-through can only be attempted from a Connected Host session.

Unrecognised Panel Received During Sign On

User may not be defined to System 21 Application Manager.
Change effected on the Host. See Logon Error Messages.

Virus Problems

See also Program Not Launching But Installed OK.

Visual Component Or Reformatted Panels Don't Appear

These are only supplied for System 21 Host applications, so will not work with non-JBA software. This is what to do:

- Check the Preferences dialogue for the SR *Path files (Pathways).
- Check the location of the SR files.

Checklists

There are checklists for AS/400, UNIX/AIX, the User and Win.ini.

AS/400

This is a suggested list of checks which you can make, in a logical order, to resolve any general installation problems:

- a) If the operating system is less than OS/400 3.1.0, get an upgrade to the supported level before taking any further steps. This may involve upgrading the router to a supported setup too.
- b) Check host libraries and folders are correctly loaded. Make sure required objects are transferred to the System portion of the library list. Refer to the Installation InfoBase file to recap details.
- c) Make sure all applications are authorised. Refer to the Installation InfoBase file.
- d) Check the configuration of your PC on the AS/400. Use the WRKCFGSTS command to make sure that the device is varied on. Check that your Router session is active.
- e) Check that the PLU name used in CSS configuration is correct. This should be the same as the system name used in your Router configuration.
- f) Check that the user profile specified in the CSSUSER statement in your Autoexec.bat file is included in the user list in the CSS configuration for your Host. Check that the host user to attach to is a valid AS/400 user profile and that the password is correct.
- g) Check the communications subsystem (QCMN) is active. Check for client/server jobs having started for your PC. If active, then look at the job log and, if necessary, change the logging level of the job. Check for any error messages and check the library list.
- h) Check the interactive subsystem (QINTER) for the Emulator session for your PC. Again, check the job log if active.
- i) Check the QSYSOPR message queue on the AS/400 for any messages regarding the failure. If there are associated jobs that started and then ended abnormally then note the Job name, user and number. Use the command WRKJOB to check the job log.
- j) Check the system history (DSPLOG), again looking for details of session failure.
- k) If the System 21 interface has run successfully, but had a failure of comms in a previous session, it may be due to one of the following:
 - Check that old jobs are not still active, using WRKCFGSTS *DEV devicename* System 21 interface sessions end in *S0 through *S7. If any are active, end them.
 - Use the command WRKUSRJOB to check for any joblogs that may have been produced for the User with which you logged on to the System 21 interface.

UNIX Or AIX

This is a suggested list of checks, in a logical order, which you can make to resolve any UNIX or AIX installation problems:

The System 21 interface does not start.

System 21 writes error logs to files prefixed `svstartup.log` in the LOGS directory off `$JBA_HOME`. Separate logs are written for each System 21 session.

Check that a log file is being produced and review its contents for any reasons why System 21 is failing to start.

If a problem occurs before errors are directed to these log files then try starting a Telnet session specifying the System 21 port number. Some output should appear on the screen when you attempt this; if not, then you should investigate your PC configuration. Check that you have entries in the services file and that the services file is in the correct directory for your PC software.

```
Telnet -p 24000 hostname
```

where *24000* is the port number and *hostname* refers to the UNIX host. This command syntax is shown for IBM TCP/IP, other software may use different parameters.

Signon screen appears on PC but unable to login

Make sure that you have set SUID on the *svsignon* program by running the following commands:

```
chown root /Release/*/exports/*/bin/svsignon  
chmod ug+s /Release/*/exports/*/bin/svsignon
```

Also check that the UserID and password are valid and can be used to access the System 21 interface from a simple character based terminal. The terminal should ideally be one that is VT220 compatible.

User

If the user is not authorised then sign on to your AS/400 as the Security Officer - QSECOFR and grant the user authority to the folder JBAS21 and the files it contains.

1. User Manual authorities. Sign on to your AS/400 using the User ID and Password which you want to use to log on to the System 21 interface. Enter the command WRKFLR and enter 5 alongside the folder JBAS21. The user should be authorised to all the files.
2. User not defined. Sign on to your AS/400 as the Security Officer - QSECOFR. Enter the command IPGCFP4/STRIPGCF select option 1 called Maintain User Profiles and then check that the user with which you will logon to System 21 exists.
3. User setup as class *USER or with LIMITCAPABILITIES. If the User has a LIMITCAPABILITIES attribute set to *YES or *PARTIAL, then it cannot be used to access the System 21 interface unless the Initial Program details are changed as specified in the JBAS21P.INI Initial program detail. Please refer to CWSINITIAL

If the user does not exist then you should add the details.

WIN.INI

The following message could display when you exit from System 21, in Windows 3.x but not Windows 95:

Application Message Queue Full

You can correct this by making changes to the INI file as follows:

1. Take a copy of the .INI file before starting the procedure.
2. Use SYSEDIT to edit Win.ini from inside Windows, or Exit Windows and use any text editor from DOS.
3. Check that by default the following two lines are present in the INI file in the [windows] section.

TypeAhead=240

DefaultQueueSize=32 *

4. This is what you can do:

- If the lines are not present then add them.
 - If the lines are already present, but the values are different to those displayed then change them to match those displayed.
 - If the lines are already present, and the values match those displayed then increase the DefaultQueueSize. If this is too low it can cause a grey screen effect in the Client Workspace Designer or Explorer Designer.
5. If you are editing in Windows, then Exit Windows because you must restart Windows for the changes to take effect.

What The Error Codes Mean

Zero	Successful
Minus value	Error. See lists below.
Positive value	TCP/IP error which hasn't been mapped to one of the generic return codes



Note: See also Client Server Configuration Utility for suggestions.

Client Server Support - Return Codes Error List

CSS Error -01 Communications Router Not Installed
CSS Error -02 Communications Router Not Started
CSS Error -03 No More File Handles Available
CSS Error -04 CPIO program not found
CSS Error -05 Unable to Load Required DLLs
CSS Error -06 No More Connection Handles Available
CSS Error -07 Invalid Scan of Table
CSS Error -08 Invalid Host Type
CSS Error -09 No FFD File Found
CSS Error -10 Invalid File
CSS Error -11 File Not Found
CSS Error -12 Invalid Program Name
CSS Error -13 Invalid Operation For File
CSS Error -14 Invalid Key Depth For File
CSS Error -15 File Already Open
CSS Error -16 Too Many Files Open
CSS Error -17 Invalid Numerics
CSS Error -18 Call Host Program Failed
CSS Error -19 Execute Host Command Failed
CSS Error -20 File Override Failed
CSS Error -21 Open Query File Failed
CSS Error -22 CPIO Program Failed
CSS Error -23 Open File Failed
CSS Error -24 File Not Open

CSS Error -25 No Record(s) Found
CSS Error -26 No More Records
CSS Error -27 Write Record Failed
CSS Error -28 No Records With Equal Key
CSS Error -29 Update Record Failed
CSS Error -30 Delete Record Failed
CSS Error -31 Locked Record
CSS Error -32 Unauthorised Access
CSS Error -33 Unauthorised Action or Operation
CSS Error -34 No Commitment Control on Host
CSS Error -35 Commit Failed
CSS Error -36 Rollback Failed
CSS Error -37 File Close Failed
CSS Error -38 Create Local File Failed
CSS Error -39 Read Failed
CSS Error -40 Invalid Parameter
CSS Error -50 Level Check Error
CSS Error -60 No Local Application List Present
CSS Error -61 Unknown Local Application
CSS Error -70 Required Local Logon Not Present
CSS Error -71 No Host User Entry
CSS Error -72 No Host User List
CSS Error -73 No Host User Entry
CSS Error -74 No User Host List
CSS Error -80 Function/Operation Not Supported
CSS Error -81 Memory Request Failed
CSS Error -90 The Connection Attempt Failed
CSS Error -97 Locking Error
CSS Error -98 Synonym Error
CSS Error -99 Communications Error

SQL - Return Codes Error List

SQL Error -0001 Not Started
SQL Error -0002 Communication Error
SQL Error -0003 Incompatible Release Levels

SQL Error -0020 Database Not Found
SQL Error -0021 Database Not Supported
SQL Error -0022 Error
SQL Error -1000 Session Not Started
SQL Error -1001 Session Already Started
SQL Error -1002 Database Not Located
SQL Error -1003 Database Not Closed
SQL Error -1004 Invalid Logon Data Area
SQL Error -1005 Invalid Cursor Data Area
SQL Error -1006 Invalid Bind Data Area
SQL Error -1007 Invalid SQL Statement
SQL Error -1008 Exec Failed
SQL Error -1009 Cursor Not Opened
SQL Error -1010 Operation Failed
SQL Error -1011 Cursor Read Only
SQL Error -1012 Column Not Found
SQL Error -1013 Datatype Mismatch
SQL Error -1014 Convert Failed
SQL Error -1015 Datatype Not Supported
SQL Error -1016 Column Does Not Exist
SQL Error -1017 Operation Not Valid Now
SQL Error -1018 Prepare Failed
SQL Error -1019 Non Select Statement
SQL Error -1020 Statement Too Long
SQL Error -1021 Exec Error
SQL Error -1022 Fetch Required
SQL Error -1023 Not Update Mode
SQL Error -1024 Fetch Failed
SQL Error -1025 Invalid Cursor
SQL Error -1026 Open Cursor Failed
SQL Error -1027 Last Row
SQL Error -1028 Maximum Cursors Opened
SQL Error -1029 Communication Error
SQL Error -1030 Incompatible Release Levels
SQL Error -1031 Parameter Markers
SQL Error -1032 Variable Not Bound

SQL Error -1033 Communication Error At Connection Time
SQL Error -1035 Maximum Connections
SQL Error -1036 Maximum Cursors
SQL Error -1037 Invalid Parameter
SQL Error -1038 Non Select Cursor
SQL Error -1039 Too Many Variable Names In Parameter List
SQL Error -1040 Too Few Variable Names In Parameter List
SQL Error -2000 Maximum Connections
SQL Error -2001 Invalid Logon Data Area
SQL Error -2002 Logon Data Area Not Found
SQL Error -2100 Invalid Cursor Data Area
SQL Error -2101 Maximum Cursors
SQL Error -2102 Cursor Data Area Not Found
SQL Error -2103 Total Cursors Opened
SQL Error -2200 Bind Variable Not Found
SQL Error -2201 Maximum Bound
SQL Error -2202 Already Bound
SQL Error -2300 DLLs Not Found
SQL Error -2301 DLLs Not Loaded
SQL Error -2302 DLLs Not Supported
SQL Error -2303 DLLs Already Loaded
SQL Error -2304 Maximum Logons
SQL Error -2305 Maximum DLL Sets Loaded
SQL Error -2306 Failed To Load DLLs
SQL Error -2401 Undefined Local User
SQL Error -2402 Invalid Client Application
SQL Error -2403 No Local Logon
SQL Error -2404 Database Name Not Defined
SQL Error -2405 Unable To Retrieve Database Info
SQL Error -9000 Invalid Parameters

UNIX CHP - Return Codes Error List

CHP Error -1001 CHPHostNotFound
CHP Error -1002 CHPTryAgain
CHP Error -1003 CHPNoRecovery

CHP Error -1004 CHPConnRefused
CHP Error -1005 CHPNetUnreached
CHP Error -1006 CHPTimedOut
CHP Error -1007 CHPNetDown
CHP Error -1008 CHPPFNoSupport
CHP Error -1009 CHPAFNoSupport
CHP Error -1010 CHPAddrInUse
CHP Error -1011 CHPAddrNotAvail
CHP Error -1012 CHPNetReset
CHP Error -1013 CHPConnAborted
CHP Error -1014 CHPConnReset
CHP Error -1015 CHPNoBufs
CHP Error -1016 CHPIsConn
CHP Error -1017 CHPNotConn
CHP Error -1018 CHPHostDown
CHP Error -1019 CHPHostUnreach
CHP Error -1020 CHPNotFound
CHP Error -1021 CHPIntr
CHP Error -1022 CHPBadF
CHP Error -1023 CHPAcces
CHP Error -1024 CHPFault
CHP Error -1025 CHPInval
CHP Error -1026 CHPMfile
CHP Error -1027 Local Logon
CHP Error -1028 Server/Host
CHP Error -1029 UserID
CHP Error -1099 CHPUnknown
CHP Error -2000 CHPNullProgram
CHP Error -2001 CHPPassword
CHP Error -2002 CHPService
CHP Error -2003 CHPPipeOpen
CHP Error -2004 CHPWriteOnSocket
CHP Error -2005 CHPInvalidType
CHP Error -2006 CHPInvalidLength
CHP Error -2007 CHPFilePermissions
CHP Error -2008 CHPPathComponent

CHP Error -2009 CHPNameTooLong

CHP Error -2010 CHPBrokenPipe

LU2 Emulator - Return Codes Error List

LU2 Error -16 to -20

LU2 Error -15 LU2_DBCS_TRAN

LU2 Error -14 LU2_ALREADY_OPEN

LU2 Error -13 LU2_INHIBITED

LU2 Error -12 LU2_NO_SESSION

LU2 Error -11 LU2_NOT_STARTED

LU2 Error -10 LU2_NO_HOST_INI

LU2 Error -9 LU2_NO_ROUTER

LU2 Error -8 LU2_NULL_STRING

LU2 Error -7 LU2_BAD_ROUTER

LU2 Error -6 LU2_BAD_CHARACTER

LU2 Error -5 LU2_QUERY_FAILED

LU2 Error -4 LU2_NO_SYSTEM

LU2 Error -3 LU2_DEFAULT_ERROR

LU2 Error -2 LU2_OPEN_ERROR

LU2 Error -1 LU2_SYSTEM_ERROR

LU2 Error 0 LU2_INFORMATIONAL

LU2 Error 1 LU2_SEND_FAILED

LU2 Error 2 LU2_DEFAULT_TRAN

LU2 Error 3 LU2_OEM_AND_FILE

Notes On LU2 Messages

There are three types of router connections:

JVW Connect to AS/400 via third party router (For
example, Client Access / NetSoft)

JVN Connect to AS/400 via TCP/IP

JVT Connect to Retrofit (UNIX or NT) via TCP/IP

Files

Config.sys File

As Config.sys is executed only when you boot the PC, you must restart your PC for any changes to take effect.

Share command

The System 21 interface requires that the program Share.exe is installed when you start your PC. The System 21 interface will check for this and offer automatic update to the User. Check that this is not currently applied by entering the command SHARE either from a Windows DOS prompt or from DOS. You should get a message back that says "SHARE already installed"

If required, take a copy of your PC's Config.sys file and insert the following line in the Config.sys file:

```
INSTALL = C:\DOS\SHARE.EXE
```

(Or you could use C:\Windows\Command\Share.exe - either file update is valid.)

The installer will update Autoexec.bat

Out of environment space

Your PC may not have sufficient environment space to hold the environment variable definitions. You can increase the environment space by including the following SHELL command in your PC's CONFIG.SYS file:

```
SHELL=C:\DOS\COMMAND.COM C:\DOS /P /E:1024
```


Autoexec.bat File

All Autoexec.bat modifications require a PC reboot for any changes to take effect.

Added SET statements

Enter the command SET either from a Windows DOS prompt or from DOS. All of the SET commands placed in the Autoexec.bat file during the installation of System 21 should now be displayed. These should include:

```
SET MSGDIR=.
SET JBAFILES=.
SET CSSXLAT=.\Jbagtab.dat
SET CSSUSER=USERID
```

The '.' represents the current directory and should be the System 21 directory. USERID is the standard local logon supplied in the CSS files as shipped.

If these statements exist in the Autoexec.bat file before installation, they will not be altered by it, unless overridden via a prompt within the installation. If the decision was taken to leave SET statements unchanged, when they need to be altered by the installation of this product, the Utilities and Applications within System 21 will look to the files in the previously setup directories for Client Server information. The Client Server Configuration Utility would use the same files and show the same data as a previously installed CSS Utility.

SET statements not being applied

This could be caused because:

- The Autoexec.bat file handles multiple configurations and the SET statements have been placed in an inappropriate part of the file. For example, if they are added after a STARTPCS statement, they will not get executed. This can be resolved either by repositioning the SET statements to BEFORE the STARTPCS, or by changing this line to CALL STARTPCS. This will force later statements in the file to be run after completion of the STARTPCS command.
- The SET statements have been placed after a command which exits the Autoexec.bat file and are not executed. For example, if they are added after a WIN statement, they will not be executed. This cannot be resolved with a CALL, as it must be the last task run in an Autoexec.bat file. The SET statements must be repositioned.

JBAS21P.INI Client Workspace Configuration File

```
;CLIENT WORKSPACE INI file - VERSION 3.5 & later...  
[JBAS21DEFAULTS]  
;JBA System 21 defaults
```

```
;The first section may be manually maintained...
```

```
Designer=0  
Language=UKB  
LanguageFile=.\JBAS21.UK  
LoadBalance=0  
LoadBalanceProg=.  
LoadBalancePath=.  
ProductName=JBA System 21  
LueInfo=0  
OverrideTitleText=0  
LueInfoCfg=.
```



Warning: The remainder of this file is maintained by the system. It should not be changed manually unless under direct supervision/direction from JBA Software Products Ltd.

```
;  
;Defaults for Host Settings  
;  
HostName=  
S21User=USERID  
IniEnvironment=0  
CssConnection=0  
Program=CWSINITIAL  
OEMCONV=1  
SessionIDLetter=J  
jqserv=24000  
;  
;Defaults for Preferences  
;  
Font=Arial Monospaced  
Colors=0,12632256,16711680,12632256,8388608,12632256,8421376,12632256,65535,12632256,128,12632256,255,12632256,0,12632256,16711680,12632256,8388608,12632256,8421376,12632256,65535,12632256,128,12632256,255,12632256  
SelectHost=1  
HostHelpOnly=0
```

```

TextOnly=1
TypeAhead=1
HideConfig=0
;
;Defaults for Paths
;
SMUPath=.\SMU
SRCPath=.\SRFILES
SRUPath=.\SRFILES
HelpPath=.\HELP
BusinessProcessPath = .\SRFILES
;
;Defaults for System Preferences
;
MaxSessions=8
Sys21Res=45
App21Res=45
FrameSize=0.9
SessSize=0.9
Recogniser=1
ClipTop=2
TitleStyle=0
RemoveStops=1
DesignFiles=1
MaxMLEs=32
Tick=50
;
;Insert [JBALUEKEYS] section for Keyboard Mapping

[ FONTS ]
;
;Section contains Fonts used by Client Workspace. It is assumed that they have ;previously
been installed
;
Arial Monospaced=ANSI
Arial Monospaced Bold=ANSI
Arial Monospaced CE=Symbol
Arial Monospaced CE Bold=Symbol
Arial Monospaced Zero-Dot=ANSI
Arial Monospaced Zero-Dot Bold=ANSI

[JBALU2]

*****
;Section contains details that may require translation...
;Ensure that a DOS editor is used for this first section
;
;

```

```
;Initial words on special panels...
;
WPM=Display Program Messages
WM=Display Messages
PE=Sign-on Information
PX=Change Password
AS=ASSIST
;Ensure that a Windows editor is used for the remainder of this section
;
;Host Message Text - message boxes on session windows
;
HMTTitle=Host Message Dialog
HMMore=&More
HMOK=&Reset
HMHelp=&Help
HMCopyright=(C) COPYRIGHT
;
;Recognition of Function Keys - soft coding of Function key - button conversion
;
FEnter=Enter
FPrint=Print
;
;Context Menu Text - right click on session window
;
XCopy=&Copy
XAbout=&About Emulator
XSelectAll=&Select All
XReset=&Reset
;
;About Box text - option off context menu
;
ATitle=About Emulator
AVersion=Version
ACoords=Coordinates
ATop=Top Row
ABottom=Bottom Row
ALeft=Left Column
ARight=Right Column
ANewMagics=New Magics
AAllMagics=All Magics
;
;Note: The Country section 'Text' fields may also be translated.
;
;END OF TRANSLATION SECTION
;*****
[Countries]
;Country details
NumCountries=16
```

```
DefaultCountry=UKB  
;There follows details of codepages and character sets]
```

```
[Hosts]
```

```
;The following sections will contain overrides setup for any Hosts configured via Client  
;Workspace  
;  
;....End of file  
;
```



Note: The Installer will make certain changes to minimise User Manual changes. Configurable settings can all be changed from within Client Workspace and should not be changed Manually unless absolutely necessary.

JBASMG.INI Explorer Configuration File



Note: This file is divided into 6 areas, in this order:

- First Manual Update Only Area
- Preferences Area
- System Configuration Area
- Second Manual Update Only Area
- Translation Area
- Client-specific Area.



Warning: Please confine your editing to those areas marked as Manual Update Only, unless under direct supervision from JBA Software Products Ltd.

The majority of the file is maintained by the system or by Preferences/Configuration settings in Explorer. It is important not to override these manually.

```
[JBAS21DEFAULTS]
;
; JBA System 21 Explorer INI file - VERSION 3.5 & later...
;
; The majority of this file is maintained by the System,
; or by Configuration/Preference Settings
; There is one area requiring TRANSLATION.
;
; All areas for MANUAL UPDATE ONLY are marked as such.
; THE REMAINDER OF THIS FILE SHOULD NOT BE CHANGED MANUALLY
; UNLESS UNDER DIRECT SUPERVISION/DIRECTION FROM
; JBA SOFTWARE PRODUCTS LTD.
;

; *****
; START OF MANUAL UPDATE ONLY AREA...
; *****
TranslationLanguage =eng
Title=JBA System 21 Explorer
ChangeGUIFont=0
JBAWebLocation=http://www.jbaworld.com
MaxSessions=8
SessionSize=0.72

;
; Defaults for Host Settings
;
```

```

Program=CWSINITIAL
SessionIDLetter=J
jqserv=24000

;Uncomment this line if using a router
;OEMCONV=1

;
; Debug Trace setting for Explorer/Session Manager
;
Trace=0

;
; Debug Trace settings for the emulator control
;
;LueDbg=22
;JvnDbg=63
;JvtDbg=63
;JvwDbg=63
;LckDbg=228

;Start the cloud
UseCloud=0

; Use to override the default prompt button bitmap
;?=c:\program files\jbasys21\BinoGray.bmp

;[LueColors]
;OutBack=000000
;InpBack=404040
;RevInpBack=00FFFF
;Red=0000FF
;Green=00FF00
;Blue=FF8000
;Turquoise=FFFF00
;White=FFFFFF
;Yellow=00FFFF
;Pink=FF00FF

; *****
; END OF MANUAL UPDATE ONLY AREA...
; *****

;
; AREA MAINTAINED BY THE PREFERENCES OPTIONS WINDOW
;

;Run QARun

```

```
;Launch Explorer on Role Menu Change
RMULaunchStatus=0
```

```
;Refresh To Do List
RequireRefresh=0
;Frequency
IntervalRefresh=10
```

```
;Function Key Text
FunctionKeyTextOnly=0
```

```
;Use Designer
Designer=0
```

```
;System Setting for Removing Stops
RemoveStops=1
```

```
;Hide the Document Viewer if there are no valid keys for the Linked Fields
HideNoKeysForLinkedFields=0
```

```
:Currency Conversion Flag
CurrencyConversion=0
```



```

ManualWindowConfiguration=0

;Show Visual components
ShowNewLooks=0

;Enable the "prompt for application server settings" button
SysSettingsButton=0

;
; Current Runtime Settings Area
;
; System 21 Explorer will set the following at runtime.
;

;Show Role Menu Designer
RMUDesigner=0

[JBALUEKEYS]
; *****
; START OF MANUAL UPDATE ONLY AREA...
; *****
; Enter below any keyboard mapping overrides (eg. SEND=CONTROL)

[LUEHOOKS]
; In the LueHooks section, all of the modules to be hooked
; (so that Text can be captured with a mouse click)
; are listed with value 1.
; All of the Classes where it is OK to just get the text
; are listed with value 2
MSVBVM50.DLL=1
MSFLXGRD.OCX=1
COMCTL32.DLL=1
ThunderRT5CommandButton=2
ThunderRT5CheckBox=2
ThunderRT5TextBox=2
ThunderRT5OptionButton=2

[ FONTS]
;
; Section contains Fonts used by Explorer. It is assumed
; that they have previously been installed...
;
Arial Monospaced=ANSI
Arial Monospaced CE=Symbol
Arial Monospaced Zero-Dot=ANSI

[JBALU2]

```

Font=10.Arial Monospaced

```
; Uncomment this for more reliable pop up recognition, provided that all
; JBA System 21 pop ups have magic numbers.
; MagicOnPopUp=1
```

```
; Set OldEvents to 1 to run Visual Components designed
; for Explorer version 1.1.3
OldEvents=0
;
; Other Visual Component related information
; Trigger a Comms Status Event when AS/400 changes state of message lite
;UpdateMsgLite=1
;
```

```
; Note that if ShowNewLooks=1, all of the following default to 0
;           ShowNewLooks=0, all of the following default to 1
; Uncomment a line to make its value independent of ShowNewLooks
; Redraw screen immediately the AS/400 tells us to
;RedrawOnRestore=1
; Show Full Screen Visual Component behind other Visual Components
;ShowBaseVC=1
; Capture System Request and Attention over Visual Components
;AllowVCSysKeys=1
; Don't display hidden messages over VCs or PopUps
;HideHiddenVCMessage=1
; Ensure data entered into Visual Component is saved
;UpdateBoth=1
; Correctly track handle to Visual Component against host screen
;UpdateHVC=1
; Remove handle to Visual Component when the Visual Component is removed
;RemoveHVC=1
;
```

```
; *****
; END OF MANUAL UPDATE ONLY AREA...
; *****
;
```

```
;
; *****
; AREA CONTAINS DETAILS THAT MAY REQUIRE TRANSLATION...
; *****
;Ensure that a Windows editor is used for translation.
```

```
;
; Initial words on special panels...
;
WPM=Display Program Messages
WM=Display Messages
PE=Sign-on Information
```

```

PX=Change Password
AS=ASSIST

;
; Host Message Text (Message boxes on Session Windows)
;
HMTTitle=Host Message Dialog
HMMore=&More
HMOk=&Reset
HMHelp=&Help
HMCopyright=(C) COPYRIGHT

;
; Recognition of Function Keys (Soft Coding of F.Key-Button conversion)
;
FEnter=Enter
FPrint=Print

;
; Context Menu Text (right click on session window)
;
XCopy=&Copy
XAbout=&About Emulator
XSelectAll=&Select All
XReset=&Reset
XWindow=&Window
XConfigure=&Configure
XReload=Re&load
XRecognition=&Recognition
XShowMagics=&Show Magics
XNoConfigure=Error Running

;
; About Box text (option off of Context menu)
;
ATitle=About Emulator
AVersion=Version
ACoords=Coordinates
ATop=Top Row
ABottom=Bottom Row
ALeft=Left Column
ARight=Right Column

;
; Note: The Country section 'Text' fields (only) may also be translated.
;

[Countries]

```

```
;  
; Country details  
;  
NumCountries=16  
DefaultCountry=UKB
```

```
Country0=AGB  
Country1=CAB  
Country2=CSB  
Country3=DMB  
Country4=FNB  
Country5=FQB  
Country6=FAB  
Country7=ICB  
Country8=ITB  
Country9=NEB  
Country10=NWB  
Country11=PRB  
Country12=SPB  
Country13=SWB  
Country14=UKB  
Country15=USB
```

```
[AGB]  
CodePage=697  
CharSet=273  
Text=Austria/Germany
```

```
[CAB]  
CodePage=697  
CharSet=260  
Text=Canadian/French
```

```
[CSB]  
CodePage=959  
CharSet=870  
Text=Czech
```

```
[DMB]  
CodePage=697  
CharSet=277  
Text=Denmark
```

```
[FNB]  
CodePage=697  
CharSet=278  
Text=Finland/Sweden
```

[FQB]
CodePage=697
CharSet=297
Text=France QWERTY

[FAB]
CodePage=697
CharSet=297
Text=France AZERTY

[ICB]
CodePage=697
CharSet=871
Text=Iceland

[ITB]
CodePage=697
CharSet=280
Text=Italy

[NEB]
CodePage=697
CharSet=037
Text=Netherlands

[NWB]
CodePage=697
CharSet=277
Text=Norway

[PRB]
CodePage=697
CharSet=037
Text=Portugal

[SPB]
CodePage=697
CharSet=284
Text=Spain

[SWB]
CodePage=697
CharSet=278
Text=Sweden

[UKB]
CodePage=697
CharSet=285

Text=United Kingdom

[USB]

CodePage=697

CharSet=037

Text=USA/Canada

;

; *****

; END OF TRANSLATION AREA

; *****

;

;

; Information added after this point is Configuration information
; specific to this Client.

; It will have been added by the System Configuration Utility

; IT SHOULD BE MAINTAINED VIA THIS UTILITY

;

;

[Currency]

GLCompanyCode=""

RateCode=""

Triangulation=0

TriangulationCurrency=""

FromCurrency=""

ToCurrency=""

Routers

Routers are a common cause for PC application problems. If a PC application it could be because of the router used as much as the application itself. To compare, try running a standard well-known popular application (word processing applications for instance) for a long time, using connections to a network. It too will suffer from Router problems.



Note: See also Gateways for information about router components.

Issues Specific To All Connection Types

These issues can be checked for and will provide some benefit:

- Many routers allow a maximum number of sessions to be set. If this is exceeded by the System 21 interface, a Communications Error message will occur.
- CMOS settings
It is essential that you disable the CMOS power saving, although normal screen savers will not cause a problem. This is because if the BIOS power saving cuts in there is only a limited time before the host terminates the connection. Also the PC will be unaware that the connection has been terminated and it will resume and report that the connection is still in place.

NetSoft Routers

Please note the following points:

- Netsoft recommend that the 32 bit Microsoft DLC driver be used in preference to the 16 bit driver for their router
- When setting up a Netsoft router configuration, with the 802.2 protocol, it is advisable to change the Advanced setting MAXDATA from 1033 to 521. This will help prevent the System 21 interface from 'hanging' with this protocol.
- Changing the following parameters in the *.RTR router configuration file chosen for use with System 21 will improve product stability: PEEKMESSAGE = 0
ACTIVATELINKTIMEOUT = 0.
- These already exist in the file, and should be amended accordingly
- The changes to TypeAhead and DefaultQueueSize in are particularly worthwhile with the NetSoft router
- It is imperative that the router is started BEFORE starting the System 21 interface. Failure to do so (that is. Allowing the System 21 interface to start the router itself) is likely to have side-affects within the application itself
- This router is prone to self-terminating its link, especially on exiting the System 21 interface. The router does this automatically on Disconnecting its own Display

sessions. It means that in order to restart the System 21 interface, the router must be reconnected first (as when using the NetSoft AS/400 display session)

- Character Translation problems. Please refer to Character Translation And Language Support and OEMCONV

Client Access For Extended DOS Router

Should a Communications Error in Session message be displayed whilst using this router to connect to a second or subsequent session, it is likely you can alter the router configuration to cure the fault.

Amend the Startpcs.bat file statement to match the following:

```
PCSWIN /B=48/z
```

It is frequently set to 16, but should be 48.

Client Access For Windows 95

IBM recommend that you install the Windows 95 Upgrade Service Pack A if using Client Access for Windows 95 product for certain versions.



*Note: To find the version number select
Settings\Control Panel from the Start menu.
Then double click on the System icon.*

This is what you do:

- If it is version 4.00.950 then install the service pack.
- If it is version 4.00.950a then the service pack is already installed.
- If it is version 4.00.950b or c or later then you have a later version of Windows 95 so do not install the service pack.

TCP/IP Protocol



If you use TCP/IP to connect to an AS/400 Host, you must install the TCP/IP licensed program (57xxTC1) on to the AS/400. Please note that this is automatically installed with Operating System OS/400 V3R7M0 or later

TCP/IP needs to locate an address for each different host to which it connects. The Operating System does this by querying a remote DNS Server and searching for the address and host. Once it has obtained the address, it can make a connection to it.

If it cannot locate the server or the address, it may take a while to respond. In addition, there is a performance overhead implied with the search.

TCP/IP has a local hosts file which is checked first, whereby the address can be found locally, and the connection then made. It is faster, but involves a change on each PC wishing to benefit from it.

TCP/IP uses a file called Hosts. If it does not exist, then local address lookup cannot be done.

A sample of this file is shipped with Windows. The sample is called Hosts.sam. This is where you can find this file:

- Windows 95 - look in the \Windows directory
- Windows NT - it is in the \WinNT\System32\Drivers\etc directory
- Windows 3.x, it is usually in the \Windows directory but sometimes in the TCPDOS\ETC directory

Host.sam looks like this, when shipped:

```
# Copyright (c) 1994 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Chicago
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#      102.54.94.97      rhino.acme.com      # source server
#      38.25.63.10      x.acme.com         # x client host
127.0.0.1      localhost
```

These are the steps you need to take to gain performance benefits with this change:



Warning: Only follow this short-cut if you are familiar with TCP/IP configuration.

Step 1

Replace the address for 127.0.0.1 LOCALHOST in Host.sam with the TCP/IP address for the Client itself, for example:

```
123.456.78.1 PCname
```

This is different for each PC.



Note: If the file Hosts already exists, then check and change that file, rather than using the sample file Hosts.sam.

Step 2

Add in more addresses, one for each host you wish to access faster. Follow the examples in the file, for example:

```
123.456.78.3 host1
```

```
123.456.78.4 host2
```

Step 3

Save the file. If the file is new then rename it from Hosts.sam (sample file) to Hosts.

TCP/IP and the Operating System will now check this file before attempting any address search. Any hosts not added to the file can still be connected to, but it will need to search in the standard manner to locate the Host TCP/IP address.



Note: Any changes to TCP/IP addresses must be reflected in this file, otherwise the subsequent connection will fail.



Note: This technique is not appropriate if dynamic TCP/IP addresses are being used.

Gateways

Gateways are router components.

Novell Netware SAA Gateway Patch

If you are using Novell Netware SAA gateway version 2.0.22 please use the SAA016 patch to make System 21 run successfully:

Other Emulators

When starting the System 21 interface you should make sure that other emulators, such as Rumba, are not running on the PC. You may start your emulator once System 21 has been started.

AS/400 Security Levels

You could encounter problems if you use anything other than Security Level 30 on your AS/400. System 21 expects a user profile and password to be entered to sign on.

Memory Problems

Running under Windows can cause memory problems on some PCs. This usually manifests itself by causing a GPF in G21AWB3D or CSPE400W. There are a number of ways to fine tune your PC.

1. Make sure that you are only running programs that are required. For example, some people run File Manager automatically as part of their start-up routine.
2. Consider removing non-mandatory settings from your Config.sys and Autoexec.bat files. For example, DOSKEY and Anti-virus programs.
3. Optimise your PC memory using MEMMAKER. This comes as part of DOS Version 6 and higher.
4. If you are using Virtual Memory, consider running without it. Although slower to run, this can cause less memory problems.
5. Use VShare.386 in Windows. This allows the local file access by System 21 and Viewfinder to run without problems. You must add the following line to C:\Windows\System.ini (remember to take a save copy first):

```
[386Enh] DEVICE=C:\WINDOWS\SYSTEM\VSHARE.386
```

Window Recognition Utility

This is a System 21 Explorer utility for fixing minor screen issues. This is where a pop up window in the application software does not appear correctly.



Note: This was originally resolved in Client Workspace by using the LUEINFO and LUEINFOCFG settings in Jbas21p.ini. The equivalent setting in jbasmg.ini for Explorer is ManualWindowConfiguration=0.

The Window Recognition utility runs the JbaLiMnt.EXE program. These details are passed to the program:

- The path to Lueinfo.cfg
- The magic numbers and co-ordinates for the problem window.

It is delivered with Explorer as part of Administration Tools. Please refer to the Tutorial for Installing The Software and the separate help text for Explorer where there is a section called Window Recognition.

Character Translation And Language Support

Different machines have different ways to represent the same character. So data retrieved from a Host machine must be modified before it is displayed on the PC. Occasionally, especially in international installations, the incorrect PC character is displayed for a given Host character. For example, a \$ is displayed instead of a £ or a ' instead of {.

If you experience this problem then you need to modify the character translation configuration. There is a utility provided with System 21 Explorer, Character Conversion, that you can use to edit the character translation tables.

The Character Conversion utility is delivered with Explorer as part of Administration Tools. Please refer to the Tutorial for Installing The Software and the separate help text for Explorer where there is a section called Character Conversion.

Options Available

The options detailed in the following sections are available for each connection protocol.

AS/400

Character translation can be done just by the router (default).

If the router translates the host code page into a MS-DOS code page, OEMCONV (OEM to ANSI Conversion) can be turned on.

For complete control, a character TranslationFile can be specified, a variety of different tables are shipped.

AS400 TCPIP

Character translation is done using shipped translation file (default).

For complete control, a character TranslationFile can be specified, a variety of different tables are shipped.

To enable existing translation files which translate the host code page into a MS-DOS code page, OEMCONV (OEM to ANSI Conversion) can be turned on.

UNIX Or NT

No translation can be done (this is the default).

For complete control, a character TranslationFile can be specified, the file NoTran.Dat is shipped as a starting point for developing custom Translation Files.

If the host happens to work in the same code page as the PC, OEMCONV (OEM to ANSI Conversion) can be turned on.

AS/400 Character Translation Using A Router

The router must be correctly configured. Determine the AS/400 and PC code pages and ensure that the router is configured for these code pages.

AS400 Character Translation Not Using A Router

This is similar to AS400 TCPIP character translation except that a Translation File must be specified somewhere in the INI file (as by default, AS400 defaults to using the routers translation functions).

AS400 TCPIP Character Translation

By default, the Translation File Jbagtab.dat is loaded. This file is found in the same directory as the System 21 INI file (for example: C:\Jbas21 for Client Workspace, C:\Program Files\JBASys21 for System 21 Explorer). The shipped Jbagtab.dat is a copy of the shipped file E037ansi.Dat.

To change the character translation, either change Jbagtab.dat, or specify a Translation File in the INI file.

If you have a translation file which converts between an AS/400 code page and an OEM code page (on the PC) and the PC has that MS-DOS OEM code page configured, then this Translation File can be used with OEMCONV turned on (this is provided for backward compatibility, in general, a Translation File should convert between the host code page and the PC font with no other intervention).

Japanese Translation

For more information please refer to the tutorial which explains how to install the Japanese version of System 21 Explorer.

Double Byte Character Set (DBCS)

Client Workspace does not support DBCS.

Explorer does support DBCS.

The first language is Japanese.

Trace

Should failures occur during the running of the System 21 interface, diagnostic trace is built into the application. Turn the trace on to produce a debug file.



Warning: Exit from the System 21 interface before you turn on the trace.

Turning On The Trace

In Windows 95

By default the trace is turned off but if you amend the following files then as soon as you open Client Workspace or Explorer the trace details are put into the trace files.

- For Client Workspace you turn trace on in Autoexec.bat
- For Explorer you turn trace on in the configuration file jbasmg.ini.

In Windows NT

Set up the trace as a system environment variable.

Trace Settings

There are trace settings for the basic process and trace settings for other components within the System 21 interface.

Basic Process

Interface	What to set	Where
Client Workspace	S21DBG=1	Autoexec.bat
Explorer	Trace=1	jbasmg.ini

Other Components

Client Workspace and Explorer

LUEDBG=282	Display issues
LUEDBG=30	Communication /memory issues
JVTDBG=8	TCP/IP Router for UNIX Table
JVNDBG=8	TCP/IP for AS/400 Table
JVWDBG=8	EHNAPPC Router to AS/400 Table

Client Workspace only

CSPDBG=2	Communication issues (CSS)
SYNDBG=2	Communication issues (SynCall)
RSQDBG=1	Communication issues (SQL)
CHPDBG=C:\Temp\CHP.DBG	Call Host Program communication issues - these are output separately to the file stated (creates new, or adds to existing)

Trace Files

Running the System 21 interface will produce a text file of the form *.DBG in the Temp directory as specified in Config.sys, that is there must be an entry in Autoexec.bat as follows:

```
SET TEMP=C:\TEMP
```

Place the Temp directory wherever you wish.

All the *.DBG statements which you have set can be active at the same time and a separate *.DBG file will be produced for every process.



Warning: Make sure that any existing debug (*.DBG) files have been deleted prior to running the System 21 interface with any of the trace or debug switches set on.

Trace Performance Details

When you set a trace it is important to know what impact it will have on performance times. JBA have four different recommendations:



Impact is minimal - **go ahead**



Impact is fairly light - **go ahead**



Impact is heavy - **take care**



Impact is excessive - **use extreme caution!**

Refer to the following tables for details:

- Emulator Trace Table
- TCP/IP Router for UNIX Table
- TCP/IP for AS/400 Table
- EHNAPPC Router to AS/400 Table



Note: To calculate the desired trace, simply add the various components together. For example: if you want a trace of 1 and 8 then set the trace value to 9.

Soft Menus



Note: you can only use a Soft Menu with the Client Workspace System 21 interface. Explorer does not have menus but has an Application Browser Area instead.

What Are Soft Menus?

These are menus which can be altered by the user and are not hard coded into the software.

Where To Find Soft Menu Files

The soft menus used by System 21 Client Workspace are stored on your PC in files with the extension .SMU and are contained within the directory set in the Client Workspace Preferences panel. The default is a \Smu subdirectory within the installed Client Workspace directory.

Size Limitations

There is a limit to the size and number of entries in a SMU file. This limit is due to a 64K restriction imposed by Microsoft Windows. If you have exceeded this limit it will manifest itself by options not running when they are selected. You should reduce the size of your menu to cure this problem.

Using The Soft Menu Maintenance Utility

This is an example for a Menu or Sub-Menu entries using a standard SMU technique:

Top Level menu: #Main Menu Text="~System"; Key=SYSTEM Text="~Applications", Key=APP

This is an example for a Sub-Menu within the current .SMU:

#System Menu Text="~Change Company"; Parent=SYSTEM etc. Text="~Change Environment"; Parent=SYSTEM etc.

This is an example for an Invocation of sub level .SMU menus used as standard:

Text="~Orders"; Parent=aaaaa; Action=Menu; Arg=A0ORDER.SMU; Tag=titlOrders - Advanced Order Entry"

The Client Workspace top level menus can initiate different types of task or action:

- Run an executable application such as Visualise, Viewfinder.
- Invoke a GUI task (a "Thick Client") such as Cash Allocation.
- Invoke an application task such as Change Company, Change Environment.

- Invoke a system task such as Work with submitted jobs.
- Open a Host or Host Emulation session.
- Action a menu keystroke such as "Enter".

Run an executable application

You can invoke any .exe. For example:

```
Text="~Visualise";Parent=APP;Action=Run, Arg="GL1094.EXE";Tag="Visualise"
```

Where the application is Visualise, the parent menu is the Applications menu (APP) defined in the SMU.

Invoke a GUI task

You can invoke any suitable GUI Task, that is a messaging .DLL coded to the appropriate JBA standards. For example:

```
Text="C~ustomer Service Desk"; Parent=APPC; Arg="CLASSNAME:GUISOP", METHODNAME:STRCSD";  
Tag="TASKCD 031000";
```

Where the parent menu is the Customer Services menu (APPC), the Class is GUISOP, and the initiating method is STRCSD in GUISOP.

The Tag is used by Client Workspace, which recognises the task type from the Tag and uses the GUI ask initiation panel (termed Panel B) and enters the Tag details to initiate the ask. The tag details are broken up into four fields:

TASK is used by Client Workspace to identify this as a task to start The next two characters (CD in this case) identify the Application The next three characters (spaces in this case) identify the environment The next two characters (03 in this case) identify the release level The last four characters (1000) identify the task within the application



Note: An entry must be present in the host file APG37, the Thick Client Control File and the application information must be defined to APG37.

Adding Internet Access To Your Soft Menus

You can use the Soft Menu maintenance utility to add access to other Windows programs. For example you can provide access to your Internet Browser from within System 21 Client Workspace. This is what you do:

- a) Double click on the Soft Menu maintenance utility icon in your JBA Utilities program group.
- b) Select Open from the File pull-down menu. Choose the SMU file to which you want to add options.
- c) Add an new option. For example:

```
Text="Netscape Navigator" Action=Run Arg="C:\NETSCAPE\NETSCAPE.EXE Common Return Codes and  
their meanings"
```

Here are some error return codes which you could get:

-2 Could not verify status with host. Check that the Router has been started and is configured correctly.

-8 Version mismatch. Check that the PC DLLs are in the correct directory and that there are no duplicates.

-18 Could not find program on AS/400. (AS/400 only). Check AS/400 library list. Check necessary objects are in System portion of library list (as detailed in the Installation InfoBase file).

42 Communications error in the Emulator. Check that the Host is still responding and that the Router is still running.

-90 Could not connect to Host. Check that there is only one copy of Ehnappc.dll on the PC hard disk. Verify that the PLU for the host is set up correctly.

-1017 Could not resolve host name to UNIX machines TCPIP port number / IP address. Check TCP/IP is started and check configuration

-32000 Application error Read the error text for more information.



Tip: You may find some less frequently encountered common return codes in the Guidelines product guide.

Keyboard Mapping

Keyboard mapping means associating the keys on the keyboard with desired host actions.

You can map keys separately for Client Workspace and Explorer by updating the relevant configuration file. Please refer to the Tutorial for more information.

Tutorial

This is how to map keys:

Step 1

Make sure that System 21 is not running on the PC. This is important because Client Workspace and Explorer update the configuration file when they are running.

Step 2

Take a copy of the configuration file and save for future reference:

- For Client Workspace, the configuration file is Jbas21p.ini. The default location is C:\Jbas21
- For Explorer, the configuration file is jbasmg.ini The default location is C:\Program Files\JBASys21

Step 3

Open the configuration file using Notepad or any other text editor.

Step 4

Add or update the [JBALUEKEYS] section. Please refer to the lists of Host Actions, PC Keys and Modifiers which you can use.

Example:

```
[JBALUEKEYS]  
SEND=CONTROL  
RESET=ENTER
```

This means that by pressing the CONTROL key you cause a SEND action at the host machine and by pressing the ENTER key you cause a RESET action at the host machine.

Example:

```
[JBALUEKEYS]  
PAGEUP=ESCAPE  
PAGEDOWN=SNAPSHOT  
HOSTHELP=NEXT+MENU
```

This means that pressing the ESC key should result in a move up the page, pressing the Print Screen key should result in a move down the page, and pressing the Alt and PAGE DOWN keys should display the HOSTHELP window.

Step 5

Save the changes and exit the configuration file.

Step 6

Restart System 21 on the PC.



Warning: The changes will not be recognised until System 21 is restarted.



Note: System 21 ignores invalid entries in the configuration file.

List Of PC Keys

These are the PC Keys which can be configured.

<u>PC Key</u>	<u>Description</u>
RETURN	Return or Enter
PRIOR	Page Up
NEXT	Page Down
ESCAPE	Esc
ADD	+ on the numeric keypad
SUBTRACT	- on the numeric keypad
SNAPSHOT	Print Screen
SCROLL	Scroll Lock
PAUSE	Pause
CONTROL	The Control Key on its own.

List Of Modifiers

These are the Modifiers which can be used with the PC keys.

<u>Modifier</u>	<u>Description</u>
+SHIFT	Either Shift Key
+CONTROL	Either Control Key
+MENU	Alt (not Alt Green - except in Asia)

Note that when these modifiers are used they must appear after the PC Key. For example:

SEND+SHIFT Represents pressing the Enter
key whilst holding down the shift
key

SHIFT+SEND This is not valid



Note: Multiple modifiers are allowed.

Combinations To Avoid

Please avoid the following key combinations as they are used by the system.

ESCAPE	+	CONTROL
ESCAPE	+	MENU
SNAPSHOT	+	CONTROL (does work on NT)
SNAPSHOT	+	SHIFT (does work on NT)
SCROLL	+	CONTROL
PAUSE	+	CONTROL

List Of Host Actions

These are the Host Actions which can be Configured.

<u>Action</u>	<u>Default Key</u>	<u>Description</u>
SEND	RETURN	Enter
PAGEUP	PRIOR	Move Page Up (called Roll Down on AS/400s)
PAGEDOWN	NEXT	Move Page Down (called Roll Up on AS/400s)
ATTN	ESC+SHIFT	Attention (AS/400 Operational Assistant Menu) - AS/400 only
RESET	CONTROL	Error Reset
FIELDEXIT	RETURN + SHIFT	Exit the field, deleting text to the right of the cursor with formatting
FIELDPLUS	ADD + SHIFT	Field Exit a numeric field and make it positive
FIELDMINUS	SUBTRACT + SHIFT	Field Exit a numeric field and make it negative
HOSTHELP	SCROLL	Get Host or PC Based Help (depends on Explorer settings)
PRINT	PAUSE + SHIFT	Host Print
SYSRQ	SNAPSHOT + MENU	System Request - AS/400 only
KATAKANA (Explorer only)	F3 + CONTROL	Toggles the display between showing Katakana and lower case English (Japanese Hosts only)
CAPTURE (Explorer only)	F1 + CONTROL	You hover the mouse over the text, press the key and the text is captured and sent to Explorer, for example for currency conversion.

Colours

You can configure colours for sessions in System 21 Explorer. The colour information is read in from the [LueColors] section of Jbasmg.ini.

RGB Colours

You specify the colours you require using RGB values.

You can make any colour by combining Red, Green and Blue in the appropriate quantities. Normally this colours are represented using the hexadecimal notation as follows:

BBGGRR

Each section ranges from 00 to FF (FF is hexadecimal for 255). For example:

- Black is 000000
- Red is 0000FF
- Green is 00FF00
- Yellow is 00FFFF
- Blue is FF0000
- Magenta is FF00FF
- Cyan is FFFF00
- White is FFFF.

You can represent any colour that can be displayed by a PC.

To find out the RGB value for any colour:

1. Access the Windows Display Properties Window, by clicking with the Right Mouse button on the Windows 95 or NT desktop, then selecting Properties.
2. Select the Appearance Panel.
3. Click on the Colour button, then click on Other.

This displays the Colour Window which shows the Red Green and Blue value in decimal notation. You can use the Windows Calculator, in scientific view, to convert this value to hexadecimal.

Setting Colours

To set the background colour of all display only, or output, fields, use:

OutBack=BBGGRR

To set the background colour of all input fields, that is field where you can enter information, use:

InpBack=BBGGRR

To set the background colour of all fields highlighted by reverse image, such as fields with invalid inputs, use:

```
RevInpBack=BBGGRR
```

You can specify which colour Explorer uses to display each of the seven AS/400 colours (Green, White, Red, Turquoise, Yellow, Pink, Blue).

For example, to specify Explorer displays AS/400 green as green, use:

```
Green=00FF00
```



Note: Explorer normally displays green fields in Blue.



Tip: If you set the background colour to red, using OutBack, then a field with the foreground set to red, would be displayed as Red on Red, therefore you could not read it. Explorer automatically corrects this problem by substituting white or black for the foreground colour. You can override these default settings using the Advanced Colour Settings.

Advanced Colour Settings

Most emulators display a Green input field in exactly the same colour as a Green output field. You can configure Explorer so that input, or entry, fields are different colours to output, or display only, fields. You can define up to four different colours for each field colour. These are:

- Output foreground colour
- Input foreground colour
- Output reverse image colour
- Input reverse image colour.



Tip: If you do not set any of these colours, Explorer uses the default.

To do this you enter up to four numbers separated by commas for each AS/400 colour.

For example:

```
Green=00FF00,008000,0000FF,FF00FF
```

This sets the green output colour to light green, the input colour to dark green, the output reverse image colour to Red, and the input reverse image colour to Pink.

Client Server Configuration Utility

Suggestions for problem solving:

- Enable the CSS Connection Flag.
- Note that the most common problems noticed in the Client Server Configuration Utility are found whilst verifying or browsing the Program/Record Access function.
- If the Browse fails after upgrading the System 21 interface then check that you have followed the supplied File Upgrade instructions correctly. Pay particular attention to the *ADD parameters or you may end up with duplicate data and the Browse will not work.
- Check the error codes.
- Check the relevant setup details for AS/400 or UNIX.
- Contact Product Support.

About Client Server Support (CSS)

The System 21 interface itself does not need CSS and therefore as a default the CSS Connection Flag is disabled.

However, some of the applications which are used within the System 21 interface do need CSS and for these applications to work then the CSS Connection Flag must be enabled. CSS will then enable these applications to connect to the host.

File Upgrade

As part of the System 21 interface (Guidelines) Host Upgrade, data from the renamed GUIDEF1 library may be copied back manually or by using one of the following conversion routines:

Routine A

Use this where the Guidelines Host libraries are only used for the System 21 interface Applications which previously existed on the Host:

```
ADDLIBLE GUIDEP1
```

```
GUIDEP1/UPDGUIDE OLDLIB(GUIDEF1OLD) MBROPT(*REPLACE)
```

Where you replace GUIDEF1OLD with the name of the saved Guidelines files library.)

Routine B

Use this where one of the following two conditions is true:

- The Guidelines Host libraries are used for other System 21 interface Applications besides those which previously existed on the Host
- System 21 interface is being added to the Host where other Guidelines applications exist:

This is an example of how to add an interface to the Host:

```
ADDLIBLE GUIDEP1
```

```
GUIDEP1/UPDGUIDE OLDLIB(GUIDEF1OLD) MBROPT(*ADD)
```

Where you replace GUIDEF1OLD with the name of the saved Guidelines files library.



Note: If problems of duplicate data arise, then the files may need tailoring manually by a DP Administrator. Duplicate data will prevent a successful Configuration Browse.

DDE

Concepts

You can use DDE to access data from an external application and take it into the System 21 interface. To do this you use the sticky fields facility where you are allocated a field in System 21 to be accessible to the 3rd party application.

Sticky Fields

To setup this facility the user must use the Designer function in the System 21 interface. The user navigates to the desired screen within The System 21 interface, then invokes the Designer function. This allows the user to identify a screen scraped field and tell the System 21 interface to save its value into a variable on the PC. The name of the variable is set by the user, and it can then be subsequently accessed either by The System 21 interface, Guidelines DLLs or by 3rd party applications via the DDE interface.

Standalone DLLs

JBA is starting to decompose business function into small standalone DLLs which can be linked together to form a complete application. These DLLs are all run from within the System 21 interface and pass their data by saving it to either global or local data areas. Only one global data area exists but there are multiple local data areas. The DDE enhancement to the System 21 interface allows external applications to share access to the local and global data areas. To support these data areas some new verbs have been added (in the enhanced message layer extension).

The System 21 interface acts as a DDE server, so any third party application which can act as a DDE client can get and set values of variables in the System 21 interface data areas, and cause the System 21 interface to execute specific functions.

JOT Implementation

JOT Verbs To Be Used Now

To write a DLL within the System 21 interface which reads or writes data from the data areas, the following JOT verbs are now to be used:

AllocateGlobalData(long gid)

This returns into gid the handle to the global id. No matter which DLL within the System 21 interface calls this, or how many times it is called, it will always return the same gid. However, another EXE will not share the same gid even if launched from within the System 21 interface.

AllocateLocalData(long lid)

This returns into lid the handle of a new data area. Every time a DLL calls this function it will be given a new lid. DLLs may share the data in a particular lid by passing it as an argument between themselves.

PutParm(long id,string name,string data)

This function puts into either the global or a local data area a variable with an associated value. If the variable already exists it will be overwritten with the new value.

string data = GetParm(long id,string name)

This function returns the value of a variable from either a local or global data area. If the variable does not exist then a MsgError will be invoked. However, the MsgError can be suppressed by preceding the GetParm() with a MsgErrorCode() function call)

Jot Verbs In The Future

In the next version of the message layer these verbs may be altered as follows:

AllocateLocalData(string name,long lid)

This returns into lid the handle of a new data area which is marked by a given name. Every time a DLL calls this function it will be given a new lid even if the name passed into the function is the same. DLLs may share the data in a particular lid by passing it as an argument between themselves.

PutParm(string name,string data, long id)

Note the placement of the id field at the end of the argument list, this is so that a new form of PutParm can be supported as follows:-

PutParm(string name,string data)

Here the data will be placed into the global data area automatically unless the name variable is of the form:

varname@localdataname where localdataname is the name of a local data area (as supplied in the revised AllocateLocalData() function).

string data = GetParm(long id,string name)

Note the placement of the id field at the end of the argument list, this is so that a new form of GetParm can be supported as follows:-

string data = GetParm(string name)

Here the data will be retrieved from the global data area automatically unless the name variable is of the form:

varname@localdataname where localdataname is the name of a local data area (as supplied in the revised AllocateLocalData() function).

More about PutParm and GetParm

Other changes are also likely as PutParm and GetParm will be the main mechanism for passing data between client programmes and Guidelines JOT generated server programs (enhancement in Guidelines version 3.4).



Note: JBA hope to restrict these to an enhanced inter DLL calling mechanism. In this mechanism

the programmer advises the system of the parameters needed by the called DLL. This information, although ignored on PC DLL to PC DLL, enables JBA to preload the server with the appropriate global/local data should the called DLL be on a server.

Failing to advise the system of passed parameters on calling a server program will not stop it running but will affect performance as the server runtime will then have to request back from the PC on a one by one basis as it encounters requests for each parameter.

Visual Basic Implementation

Example Code Module

The following is a code module using VisualBasic in Word V6. Other DDE clients have slightly different formats for their commands but they are all basically the same.

```
Sub MAIN
c1 = DDEInitiate("JBAS21", "SalesOrder") ' connects to the System 21 interface, 2nd
argument 'SalesOrder' is currently ignored.
DDEExecute c1, "EXEC=d:\ngapps\SelCust.exe" ' executes a standalone exe
DDEExecute c1, "DYND=s21insp::invoke" ' executes a function 'invoke' in DLL 's21insp'
DDEPoke(c1, "Customer", "Steel Pressings") ' save a parameter in the global data area
DDEPoke(c1, "Customer@sop", "Needle Industries") ' save a parameter in the local data area
called 'sop'. The local data area will be created if it doesn't exist
DDEPoke(c1, "OrderValue", "100.99") ' save a parameter in the global data area
customer$ = DDERequest$(c1, "Customer") ' retrieve a value from the global data area, the
returned string is empty if the parameter is not found.
OrderValue$ = DDERequest$(c1, "OrderValue") ' retrieve another value from the global data
area, the returned string is empty if the parameter is not found.
lcustomer$ = DDERequest$(c1, "Customer@sop") ' retrieve value from the local data area,
called 'sop' the returned string is empty if the parameter is not found or the data area
does not exist.
DDETerminate(c1) ' terminates the connection to the System 21 interface
Insert Customer$ ' puts the retrieved customer into the current document
Insert OrderValue$ ' puts the retrieved order value into the current document
End Sub
.
```

Future Changes To The System 21 Interface DDE Implementation

Limitation

Note that when calling the DDEExecute() method, you may invoke a task in the System 21 interface that takes a considerable length of time. You may even cause a user input dialogue from the System 21 interface to pop up in the middle of your Word application.

Unfortunately Visual Basic expects the DDEExecute task to be completed in a short time (approx. 15 seconds) and errors if the System 21 interface does not acknowledge that the task has been completed.

To avoid this problem when calling a DDEExecute, the System 21 interface currently acknowledges back to Visual Basic immediately and then carries out the requested command. This makes it possible for the Visual Basic program to continue without the operation invoked by DDEExecute having to be completed - in other words VisualBasic carries on executing in parallel with the invoked operation.

JBA will address this limitation shortly by providing a DDE method of checking if the System 21 interface command has completed from VisualBasic.



Tip: Currently you could crudely achieve this by using a parameter in the global data area as a flag.

Sticky Fields

The use of sticky fields with the Client Workspace Designer or Explorer Designer has so far been limited to transferring information between LU2 sessions. Reading or writing data to or from sticky fields could not be done either from the System 21 interface DLLs or from external applications. As of 3.4.0d, all sticky fields can be accessed through a local data area with the reserved name of CWD_DesFld (the System 21 interface Data Area, Designer Fields).

For example:

```
PutParm("ItemCode@CWD_DesFld", "12345")  
Item = GetParm("ItemCode@CWD_DesFld")
```

The parameter names in CWD_DesFld are the same as are specified in the Designer function where fields are allocated a variable name.

The currently selected company and environment will also be available through the global data area with field names CWG_Company and CWG_Environment (CWG = Client Workspace Global).

For example:

```
comp = GetParm("CWG_Company")
```



*Note: Currently you would have to do
GetParm(gid,"CWG_Parm")*

It is also planned that Version 3.4.0d of the System 21 interface will also support both pre-execution and post-execution of DLLs. Currently a DLL can only be called before the LU2 screen it has been attached to is displayed, not after the user has dismissed it.